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1ST INTERNATIONAL CONFERENCE ON INTELLIGENT SIGNAL PROCESSING AND EFFECTIVE COMMUNICATION TECHNOLOGIES (INSPECT-2024)

ABV-IIITM Gwalior, India, 07-08 December 2024

ABOUT INSPECT-2024

Welcome to the 2024 International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT-2024), to be held in December 2024 at ABV-IIITM Gwalior, India. The conference is the premier forum for presenting new advances and research results in varied disciplines and their associated applications in all aspects of Communication Technology, Signal Processing Informatics, and Machine Learning that consider improving and developing quality of human life and better living on earth. In addition to the technical sessions, there will be invited sessions, panel sessions, and keynote addresses. The aim of the conference is to stimulate interaction and convergent among researchers active in the areas of Communication Technology, Signal Processing Informatics, and Machine Learning. The conference will be sponsored by the IEEE MP Section.

ABOUT ABV-IIITM GWALIOR

Established in 1997, with a financial outlay of Rs. 61.0 Crores (the US \$14 million) for the first five years, the institute serves as a premier institution in the field of IT and management. Over the years the institute has been providing highly respected Techno-Managers to the IT industry doing various techno-functional roles with elan. The institute was accorded the status of deemed university in the year 2001. The institute campus is spread over 160 acres (0.65 sq. km) amidst a picturesque landscape in the foothills of the Gwalior Fort. The campus is self-contained and equipped with all the state-of-the-art facilities necessary for the all-round development of the student's personalities.

The Department of Electrical and Electronics Engineering is one of the major departments being created in 2022 to achieve the mission and vision of the institute. The Department is currently in its development phase and engaged in teaching, research and development in the areas of Wireless communication, multimedia signal processing, machine learning, power systems, VLSI, IoT devices, autonomous control systems, high-frequency devices, and communication systems. Department of Electrical and Electronics Engineering is organizing INSPECT-2024.

CONFERENCE TRACKS

INSPECT-2024 welcomes high-quality, original, and previously unpublished submissions in computer vision and machine intelligence. Topics of interests for the INSPECT-2024 conference include, but not limited to, the following:

Track-1: Signal & Image Processing

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| • Array Signal Processing | • Image Compression | • Sparse Signal Processing |
| • Nonlinear Signal Processing | • Pattern Recognition | • Hyperspectral Imaging |
| • Adaptive Signal Processing | • Computer Vision | • Geospatial Image Processing |
| • Audio Signal Processing | • Medical Image Processing | • Biometric Signal and Image Processing |
| • Speech Processing. | • Biomedical Signal Processing | • Deep Learning for Signal Processing |
| • Digital Signal Processing | • Remote Sensing | • Real-Time Signal and Image Processing |
| • Image Enhancement and Restoration | • Multimedia Signal Processing | • Acoustic Sensor Network |

Track-2: Communication Systems & Networking

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| • Wireless Communication Systems | • Information Theory | • Network Virtualization |
| • 5G and Beyond | • Cloud Computing | • Quantum Communication |
| • Satellite Communication Systems | • Network Protocols | • Network Performance Optimization |
| • Network Security | • Communication Hardware Design | • Green Networking Technologies |
| • Cognitive Radio Networks | • Social Networks Analysis | • Deep Learning for communication |
| • Software-Defined Networking (SDN) | • Multimedia Communication Systems | • Edge Computing |

Track-3: IoT, Smart Grid, Robotics & Control Systems

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| • Internet of Things (IoT) | • Energy Management Systems | • Renewable Energy Integration |
| • Smart Grid Technologies | • Sensor Networks | • Smart Cities Infrastructure |
| • Industrial Automation | • Autonomous Vehicles | • Home Automation |
| • Robotics and Automation | • Human-Robot Interaction | • Intelligent Transportation Systems |
| • Control Systems Engineering | • Distributed Control Systems | • Swarm Robotics, Wearable Technology |
| • Cyber-Physical Systems | • Embedded Systems Design | • Intelligent Robotics and Control |

Track-4: RF, Microwave & Optical Communication

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| • Smart Radio Frequency Engineering | • Optical Fiber Communication | • RF Circuit Design |
| • Microwave Engineering | • Free-Space Optical Communication | • Optical Networks |
| • Antenna Design | • Microwave Photonics | • Microwave Remote Sensing |
| • Radar Systems | • RF MEMS | • Terahertz Communication |
| • Millimeter-Wave Communication | • Optical Switching and Routing | • Photonic Integrated Circuits |

Track-5: VLSI for Signal Processing & Communication

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| • Digital VLSI Design | • VLSI CAD (Computer-Aided Design) | • FPGA Design |
| • Analog VLSI Design | • High-Level Synthesis | • VLSI Testing and Verification |
| • Mixed-Signal VLSI Design | • Hardware Acceleration | • SoC (System-on-Chip) Design |
| • Low-Power VLSI Design | • VLSI Design for Signal Processing | • Advanced CMOS Technologies |
| • ASIC Design | • Hardware Security | • Clock and Power Distribution Network |
| • Neuromorphic Engineering. | • Reconfigurable Computing | • VLSI Architectures for Communication |

IMPORTANT DATES

Call for Paper	15 April 2024
Submission Deadline	31 July 2024
Notification of Acceptance	10 September 2024
Camera Ready Submission Deadline	25 September 2024
Early Bird Registration	10 October 2024
Registration Close	30 October 2024
INSPECT-2024	07-08 December 2024

SUBMISSION GUIDELINES

Submission Guidelines Prospective authors are invited to submit full-length original research papers. In submitting a manuscript to INSPECT-2024, the authors acknowledge that no paper substantially similar in content has been or will be submitted to another journal, conference, or workshop during the review period. In such a case, the paper will be rejected without review. Papers must be electronically submitted without exception before the deadline expires. The paper must be at most 6 pages in length (including all text, figures, and references). However, authors can add up to 2 extra pages with the appropriate fee payment per page. The manuscript must be submitted in pdf format only, and the file size of your manuscript should not exceed 10 MB. INSPECT-2024 follows a single-blind review process. Authors must use the manuscript template as follows: Overleaf, LaTeX, or Word. The manuscript has to be uploaded online at the INSPECT-2024 using the Microsoft CMT paper submission portal.